UNIVERSITI MALAYSIA TERENGGANU

ASSIGNMENT
TUGASAN

SEMESTER II 2025/2026 SESSION (DEGREE PROGRAMME)
SEMESTER II SESI 2025/2026 (SARJANA MUDA)

TOPIC: TIME SERIES METHODS (EVEN-NUMBERED GROUPS ONLY)

❖ General Requirement (IMPORTANT).

- Search and select datasets independently from the internet (e.g., Kaggle, UCI Repository, World Bank, Yahoo Finance).
- Use real-world business or economic data.
- Clearly state: Data source, Data description, Software (R / Python / Excel)

❖ Task Description: Students are required to perform a complete time series analysis, starting from data exploration to ARIMA forecasting model development.

- Dataset Requirement
  i) A time variable (date/time)
  ii) Minimum 50 observations (recommended ≥100)

- Required Analysis
  i) Time Series Methods

- Report Structure
  i) Introduction
  ii) Data Description
  iii) Methodology
  iv) Results and Discussion
  v) Conclusion
  vi) References

TOPIC: DATA REDUCTION AND BIG DATA (FOR ODD-NUMBERED GROUPS ONLY)

❖ General Requirement (IMPORTANT)
- Search and select datasets independently from the internet (e.g., Kaggle, UCI Repository, World Bank, Yahoo Finance).
- Use real-world business or economic data.
- Clearly state: Data source, Data description, Software (R / Python / Excel)

❖ Task Description: Students are required to apply data reduction techniques on a multivariate dataset.

- Dataset Requirement:
  i) Must contain at least 5 variables.
  ii) Preferably large dataset (big data context)

- Required Analysis:
  i) Data Exploration
  ii) Data Reduction Technique

- Report Structure
  vii) Introduction
  viii) Data Description
  ix) Methodology
  x) Results and Discussion
  xi) Conclusion
  xii) References

DASHBOARD DEVELOPMENT REQUIREMENTS

- Students are required to design and develop a dashboard using dataset that have been selected.
- The dashboard must:
  - Clearly visualize key findings from the analysis
  - Include appropriate charts such as.
  - Be well-organized, interactive (if possible), and easy to interpret

TOPIC: Volatility Modelling Technique (Literature-Based Task)

- Students are required to evaluate and understand the application of volatility models through analysis of a published research article.

- Task Description: Each group is required to complete the following:
  i) Article Selection
  Select ONE (1) research article related to volatility modelling technique. The article must:
    - Published in a journal or conference proceeding (Scopus/Wos)
    - Within the year 2022 – 2026

  ii) Article Review. Students must clearly explain the following components:
  (a) Background of the Study
  Research problem
  Importance of volatility modelling in the study
  (b) Data Description
  Type of data (e.g., stock prices, exchange rates, commodities)
  Time period
  Data source
  (c) Model Description
  Explain the volatility model used
  Justify why the model was chosen
  (d) Results and Findings
  Key findings
  Interpretation of volatility (e.g., volatility clustering, leverage effect)
  (e) Strengths and Limitations
  Advantages of the model
  Limitations of the study

GROUP PRESENTATION (20%): CLO2, PLO3 (PRACTICAL SKILLS), CLO3, PLO6 (DIGITAL SKILLS)

Criteria	Inadequate (1)	Needs Improvement (2)	Fair (3)	Good (4)	Excellent (5)	Weightage
Selection of Tools & Programming (CLO2)	No clear tool used	Inappropriate tool selection	Basic tool used but not justified	Appropriate tool selected with some justification	Appropriate and well-justified tool aligned with analysis	X1
Dashboard & Visualization (CLO3)	No dashboard	Poor or unclear visuals	Basic visualization	Clear and relevant dashboard	Interactive, well-designed, and insightful dashboard	X1
Implementation of Methods (CLO3)	No implementation shown	Incorrect implementation	Basic implementation with errors	Correct implementation with minor issues	Accurate and complete implementation of methods.	X1
Interpretation of Results (CLO3)	No interpretation	Weak interpretation	Basic explanation	Clear interpretation	Insightful interpretation with business relevance	X1
Communication & Delivery (CLO3)	Unclear and disorganized	Difficult to follow	Some clarity	Clear and structured	Confident, engaging, and professional	X1
					TOTAL MARKS	**25**

ASSIGNMENT REPORT (10%): CLO2, PLO3 (PRACTICAL SKILLS)

Criteria	Inadequate (1)	Needs Improvement (2)	Fair (3)	Good (4)	Excellent (5)	Weightage
Selection of Dataset & Relevance	Dataset not relevant / not business-related	Weak relevance	Acceptable dataset	Relevant dataset	Highly relevant dataset with strong business context	**X1**
Selection of Programming Tool	No tool stated	Inappropriate tool	Basic tool used	Appropriate tool selected	Appropriate and well-justified tool	**X1**
Selection of Method / Technique	No method selected	Incorrect method	Basic method used	Appropriate method selected	Well-justified selection aligned with data and objective	**X1**
Justification of Choices (Tool & Method)	No justification	Weak justification	Basic explanation	Clear justification	Strong and critical justification based on data characteristics	**X1**
					TOTAL MARKS	**20**